UNIT 3 · STANDARD 6.RP.3

Equivalent Ratios

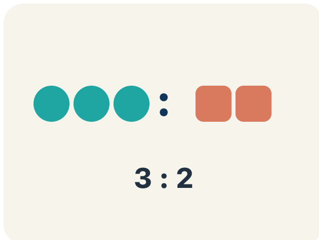
Lesson 3-4

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

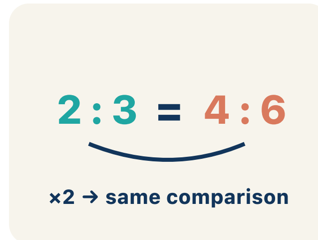
Picture first, then the word, then a plain-language meaning. Say each word out loud.



3 red apples : 5 green apples

Ratio

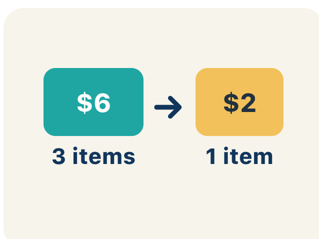
A way to compare two amounts.



2:3 and 4:6 both mean 2 out of every 3, so 2/3 = 4/6

Equivalent ratios

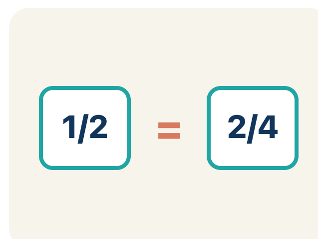
Two ratios that mean the same thing.



60 miles per 1 hour

Rate

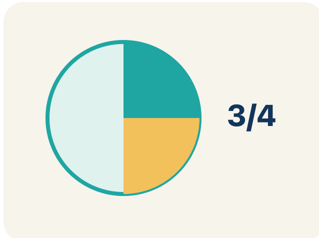
A ratio comparing two amounts with different units, like miles per hour.



2/3 = 4/6

Proportion

A math sentence saying two ratios are equal.



12:18 → divide both by 6 → 2:3

Simplify

To make a ratio smaller while keeping the same comparison.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find and check equivalent ratios using multiplication and division.

1️⃣ Notice

Chef Montoya is preparing her famous tomato soup for a school banquet.

2️⃣ Key idea

I can find and check equivalent ratios using multiplication and division.

3️⃣ Words I'll use

Ratio

Equivalent ratios **Rate**

Proportion **Simplify**

🤔 Think About It

- What quantities are being compared in the recipe?
- What stays the same when you multiply both ingredients by the same number?
- How many times bigger is 32 than 8?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Which ratio is equivalent to 2:5?

A. 4:10

B. 3:6

C. 2:10

D. 5:2

Step 1 Multiply both parts by 2: 2×2 = 4 and 5×2 = 10, so 4:10 is equivalent to 2:5.

✅ **Answer:** A. 4:10

🤝 We do — together

Solve this one with your class using the same steps.

Problem: A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

A. 2:3

B. 8:10

C. 4:3

D. 6:4

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ You do — your turn

Now try one on your own. Show every step.

Problem: Which ratio is NOT equivalent to 6:9?

A. 3:5

B. 2:3

C. 12:18

D. 18:27

Show your work:

____

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Montoya's soup serves 8 people, but she needs to serve 32. How many times bigger is the recipe, and what must she do to BOTH ingredients to keep the flavor?

Level 1 support

Start here: 32 is 4 times 8, so she multiplies both ingredients by 4.

💡 **Need a hint?**

Sentence starters (optional)

32 is ____ times bigger than 8, so I multiply both by ____.

32 es ____ veces más grande que 8, así que multiplico ambos por ____.

The flavor stays the same because ____.

El sabor se mantiene igual porque ____.

Word bank: Equivalent ratios | Proportion | Rate

Listen for: *A strong answer finds the scale factor 4 (32 ÷ 8) and explains both ingredients must be multiplied by 4 to make equivalent ratios.*

Level 2

Chef Montoya now needs to scale DOWN to serve only 4 people. Explain what scale factor she uses and why dividing keeps the ratios equivalent.

- To serve 4 people she divides both ingredients by ____ because ____.
- Dividing keeps equivalent ratios because ____.

EXPLORE

Looking at the tomatoes-to-broth table (3:2, 6:4, 9:6, 12:8), how can you PROVE these ratios are all equivalent?

Level 1 support

Start here: Each row is 3:2 multiplied by the same number, so they all simplify to 3:2.

💡 **Need a hint?**

Sentence starters (optional)

These ratios are equivalent because each is ___ times ___.

Estas razones son equivalentes porque cada una es ___ veces ___.

I checked by ___ both numbers.

Lo comprobé al ___ ambos números.

Word bank: Equivalent ratios | Simplify | Proportion | Rate

Listen for: *Listen for proof by simplifying (all reduce to 3:2) or scaling (each multiplied by the same factor), not just 'they look similar.'*

Level 2

A student says 6:4 and 9:6 are equivalent because 'you add 3 and 2 each time.' Critique this reasoning even though the rows happen to be correct.

- Adding 3 and 2 makes correct rows here, but it does NOT prove ___.
- The real reason they are equivalent is that they ___ to the same ratio.

CONNECT

The smoothie shop uses 2 cups of strawberries for every 3 cups of yogurt and needs 15 cups of yogurt. How is this the same as scaling Chef Montoya's soup?

Level 1 support

Start here: Both scale a 2-quantity recipe with equivalent ratios.

💡 **Need a hint?**

Sentence starters (optional)

This is like our ratio work because ___ and ___ are related by ___.

Esto es como nuestro trabajo con razones porque ___ y ___ se relacionan por ___.

15 cups of yogurt is ___ times 3, so I need ___ cups of strawberries.

15 tazas de yogur son ___ veces 3, así que necesito ___ tazas de fresas.

Word bank: Equivalent ratios | Proportion | Simplify | Rate

Listen for: *A strong answer uses the scale factor 5 (15 ÷ 3) to find 10 cups of strawberries and names this as making an equivalent ratio / proportion.*

Level 2

Set up this smoothie situation as a proportion (2/3 = ?/15) and explain how cross-multiplying confirms your strawberry amount.

- The proportion is 2/3 = ___/15, and cross-multiplying gives ___.
- Cross-multiplying works because equivalent ratios always ___.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words equivalent ratios, rate, proportion, and simplify.

1. Kernel Sentence

subject + verb

Model: Ratio is a way to compare two amounts.
Razón es una manera de comparar dos cantidades.

Write a kernel sentence about ratio. Use a subject and a verb.
Escribe una oración base sobre razón. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Ratio matters in math
Razón importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Ratio matters in math because ____.
Razón importa en matemáticas porque ____.

____

but *pero*

Ratio matters in math, but ____.
Razón importa en matemáticas, pero ____.

____

so *entonces*

Ratio matters in math, so ____.
Razón importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about ratio.
Di un hecho verdadero sobre ratio.

Ratio ____.

Question
Pregunta

Ask a question about ratio.
Haz una pregunta sobre ratio.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about ratio.
Muestra entusiasmo sobre ratio.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with ratio.
Dile a un compañero qué hacer con ratio.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

32 is ____ times bigger than 8, so I multiply both by ____.
32 es ____ veces más grande que 8, así que multiplico ambos por ____.

The flavor stays the same because ____.
El sabor se mantiene igual porque ____.

These ratios are equivalent because each is ____ times ____.
Estas razones son equivalentes porque cada una es ____ veces ____.

Try It

Solve on your own. Check the answer key when you are done.

1. If 12:18 is written in simplest form, what is it?

A. 2:3
B. 6:9
C. 3:2
D. 4:6

Show your work:

2. A paint mix uses 3:5 blue to white. To keep the same color with 15 parts white, how much blue?

A. 9
B. 12
C. 5
D. 8

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A store sells 3 pens for $2. Another store sells 9 pens for $6. Are these equivalent ratios? If so, prove it using at least two different methods (simplifying, scaling, or cross-multiplying).

Show your work:

Reflect — Exit Ticket

A store sells 5 notebooks for $8. At this rate, how much would 15 notebooks cost?

A. $24

B. $18

C. $20

D. $40

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 2:3 *— Divide both by 2: 4÷2 = 2 and 6÷2 = 3. The ratio 2:3 is equivalent to 4:6.*
2. **You Try (notes):** A. 3:5 *— 6:9 simplifies to 2:3. Check each: 2:3 ✓, 12:18 = 2:3 ✓, 18:27 = 2:3 ✓. But 3:5 does not simplify to 2:3.*
3. **Try It 1:** A. 2:3 *— 12:18 divides by 6 to give 2:3.*
4. **Try It 2:** A. 9 *— 15 white is 3 times 5, so blue is 3 times 3 = 9.*
5. **Exit Ticket:** A. $24 *— 15 is 3 times 5, so the cost is 3 × $8 = $24. The ratio 5:8 scales to 15:24.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Ratio is a way to compare two amounts.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).